

MATLAB LAB-4

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SECTION-CSE02
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To find the inverse of Matrix Using Gauss jordan Elimination

```
clc
clear all
close all

A=[5,12,3;4,10,13;7,14,2];
I=[1,0,0;0,1,0;0,0,1];

Aug=[A I];

Aug(1,:)=Aug(1,+)/5;
Aug(2,:)=Aug(2,)-4*Aug(1,);
Aug(3,:)=Aug(3,)-7*Aug(1,);
Aug(2,:)=(5/2)*Aug(2,);
Aug(1,:)=Aug(1,)-((12/5)*Aug(2,));
Aug(3,:)=Aug(3,)+((14/5)*Aug(2,));
Aug(3,:)=Aug(3,)/72;
Aug(1,:)=Aug(1,)+(63*Aug(1,));
Aug(2,:)=Aug(2,)-((53/2)*Aug(3,));

>> Aug

Aug =

1.0000    0.0000    0.0000   -1.1250    0.1250    0.8750
0.0000    1.0000    0.0000    0.5764   -0.0764   -0.3681
0.0000    0.0000    1.0000   -0.0972    0.0972    0.0139
```

Using For while loop to find the inverse matrix of the following given Matrix.

```
clc
clear all
close all

A=[5,12,3;4,10,13;7,14,2];
n=3;
I=eye(n);

Aug=[A I];

for i=1:n
    value=Aug(i,i);
    Aug(i,:)=Aug(i,+)/value;
```

```
for row = 1:n
    if row~=i
        multiplier = Aug(row, i);
        Aug(row, :) = Aug(row, :) - (multiplier * Aug(i, :));
    end
end
end
```

```
Inverse = Aug(:, n+1:end);
disp('Final Inverse Matrix:');
disp(Inverse);
```

Result:

Final Inverse Matrix:

```
-1.1250  0.1250  0.8750
 0.5764 -0.0764 -0.3681
-0.0972  0.0972  0.0139
```